

Orchestrating Value Co-Creation in Digital Education Using the TISE-VALORIZE Framework: From Knowing to Becoming

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12 Abstract

13 AI can automate technical activities, but it can't teach skills that only people have. This research
14 introduces the Triune-Intelligence Smart Engineering and VALORIZE framework, a conceptual
15 advancement that integrates human, artificial, and natural intelligence via a cyclical "perceive-
16 understand-decide-act-learn" cognitive cycle. Three keyways were employed to put the framework into
17 action: systematic knowledge mapping to assist specialists construct their schemas, professional
18 simulations that treat the classroom like a job, and a knowledge marketplace where student-created
19 artifacts contribute value to the community. To test this strategy, the framework was compared to
20 conventional instruction in an initial study of three groups of undergraduate engineering students.
21 Statistical results show that the experimental group did far better in school and had a much smaller
22 range of scores than the control groups. These results indicate that the framework promotes more equal
23 and consistent learning outcomes by alleviating unnecessary cognitive burden and addressing
24 psychological demands for autonomy and relatedness. This study presents a scalable educational
25 framework designed to equip graduates for professional settings characterized by the growing
26 automation of regular cognitive tasks.

27 1 Introduction

28 Engineering education is shifting from a content-based approach to competency development
29 (Charytanowicz, 2023). Other empirical studies indicate that the disparity between education and
30 industry endures, even among high-achieving graduates: analyses of software engineering practices
31 consistently reveal a misalignment between the focus of higher education programs and the
32 requirements of practitioners in professional settings, including both technical and professional aspects
33 (Garousi, Giray, and Tuzun, 2019). To build a long-lasting smart campus ecosystem, many tools and
34 frameworks are being created to test preparedness and include digital transformation (R. V. Imbar *et*
35 *al.*, 2022; Imbar *et al.*, 2025; Johan, Langi, and Imbar, 2025).

There has always been a continuous narrative in engineering education. Students study the basics, figure out how to tackle frequent situations, and test their skills with tests. High grades showed that you were ready to work in the field. But this approach included an unspoken assumption: that the mental effort needed to solve engineering issues was the same as the mental labor needed to be an engineer. This assumption is no longer valid.

To believe that something isn't real, you need to be able to ignore a lot of data and put information together in a manner that makes sense in a brain structure called schemas (Sweller, Van Merriënboer, and Paas, 1998). One important thing about working memory, which is a mental region where new knowledge is kept, is that it can only contain roughly 3–7 bits of information at a time. If you don't erase them, they will evaporate after 15–30 seconds (Miller, 1956). Students will have trouble learning and developing their abilities if the environment has too much information in it. Cognitive Load Theory (CLT) offers a solution: instructional design should minimize extraneous cognitive load to optimize working memory for effective learning termed 'germane load' by CLT researchers. CLT serves as a cognitive framework for crafting educational experiences that cultivate robust skills (Robins, 2022). It must alleviate superfluous cognitive burden to liberate working memory for authentic learning, and it must structure teaching to facilitate the development of meaningful, practical knowledge frameworks that embody professional competence. It must also do this in a manner that respects autonomy, competence, and relatedness that keep people interested. CLT posits that the complexity of a topic correlates with the interaction of its components. Darejeh, Marcus, and Sweller (2021) examined the impact of narrative style on cognitive load during the acquisition of a difficult productivity software program. The researchers used two categories of materials, distinguished by low and high element interaction, according to each story style.

We examined a primary hypothesis: can instruction developed in accordance with the TISE-VALORIZE principles, which integrate professional simulations, dual knowledge maps, value-centered evaluation, and AI-driven customization, provide significantly superior outcomes compared to traditional teaching methodologies? The outcomes we aimed for included four aspects: increased intrinsic motivation in students, enhanced professional identity development, the capacity to apply learning in diverse circumstances, and the generation of genuine community values. We also made sure that the new method didn't affect the old ones, so the test results kept the same or even became better. The secondary hypothesis in this study focusses on five primary areas. First, we looked at whether students who used the VALORIZE framework built stronger and more connected knowledge schemas (measured by the complexity and interconnectedness of knowledge maps) and reported less cognitive load during learning activities than a control group. Second, in terms of intrinsic motivation and autonomy, we inquired whether involvement in a knowledge marketplace and professional simulation, with a clear focus on freedom of choice and control over the learning process, enhanced students' intrinsic motivation and sense of learning agency relative to a control group. Third, in terms of professional identity and ethical reasoning, students who learn in a VALORIZE environment are expected to demonstrate a more robust professional identity, characterized by enhanced professional self-efficacy, ethical reasoning abilities, and metacognitive awareness, in comparison to their counterparts engaged in conventional learning. Fourth, we wanted to see whether students who learned about VALORIZE could apply what they learned to solve challenges that weren't planned or taught explicitly. If they could do that, it would suggest they had learned more and that what they had learned would be easier to use in other situations. Fifth, in terms of value co-creation, it is assessed if student creations (knowledge maps, issue solutions) in the VALORIZE environment have enough quality and utility for reuse and further development by peers and future generations, hence generating genuine community values.

The theoretical framework delineated below aims to actualize this possibility. It is based on well-known learning theories (Cognitive Load Theory (CLT), Self-Determination Theory (SDT), and research on expertise) and considers the realities of modern technology (AI integration, digital tools, and participatory design). Its goal is to create professionals who can lead and make a difference in a world that is changing quickly.

2 Theoretical Foundations: Merging Cognitive Science and Pedagogy

2.1 Fundamental Ideas About Education

The framework is based on several interconnected ideas that are meant to meet the needs of the AI future. First, Value-Oriented Education (VOE). In traditional education, teachers teach technical information, and students learn it. VOE alters this goal. Higher education should teach how to be good people, how to be professionals, and how to give back to society. Students learn how to develop code and why technological choices might hurt people or the environment, for example. Second, Value Co-Creation (VCC). This word originates from marketing and service writing and talks about how businesses and customers work together to produce value. In education, VCC means that students don't simply passively take in knowledge; they actively contribute create learning experiences that matter. They collaborate with educators and technology to develop resources such as knowledge maps, problem-solving tools, and educational aids for their classmates and future learners. In this way, students become co-creators instead than merely consumers of knowledge (Prahalad and Ramaswamy, 2004; Grönroos and Voima, 2014; Robayo-Pinzon *et al.*, 2023; Jamil, O'Connor and Shelton, 2025).

Knowledge Management (KM) is a broad way for an organization to gather, develop, protect, share, and use knowledge gained from its experiences (Mohammed, 2021; Deev and Finogeev, 2023; Fernández-Nieto *et al.*, 2024). It works well with the focus of VOE and VCC. Collaborative Learning (CL) is a strategy in which a small group of students work together to achieve a shared objective or complete an assignment (Robinson, Kilgore and Warren, 2017; Redmond *et al.*, 2018; Sidi, Blau and Shamir-Inbal, 2022; Tan, 2022; Ouyang, Xu and Cukurova, 2023; Ramadevi *et al.*, 2023). Lastly, Personalized Learning (PL) is a method of teaching that changes based on how fast students learn and how lessons are given to help them achieve their best (Labib, Canós, and Penadés, 2017; Deev and Finogeev, 2023; Apirating, 2025).

2.2 Psychological and Cognitive Foundations

CKMS-SE combines old ways of thinking about psychology and cognition:

- Self-Determination Theory (SDT) explains that individuals are intrinsically driven when their demands for autonomy, competence, and relatedness are satisfied (Weydner-volkmann and Bär, 2024; Annamalai and Nasor, 2025).
- Cognitive Load Theory (CLT) emphasizes that successful learning is facilitated by reducing irrelevant cognitive burden (extrinsic load) and enhancing cognitive load that aids the learning process (Alsaffar, Alfayly, and Ali, 2022; Apirating, 2025).
- People that have similar aims, interests, and skill levels make up Communities of Practice (CoP). They acquire information collaboratively by active participation and contribution as a cohesive unit (Tsang and Tsui, 2017; Taft, 2019; Fernández-Nieto *et al.*, 2024).

2.2.1 Self-Determination Theory and Intrinsic Motivation

CLT elucidates the cognitive frameworks integral to the learning process, while SDT emphasizes the motivating factors that sustain student engagement and facilitate profound learning (Engel *et al.*, 2023; Ninghardjanti *et al.*, 2025). SDT says that human motivation exists on a scale. At one end is external regulation, where people do things because they feel pressure or get benefits from outside sources. At the other end is integrated regulation, where people do things that are in line with their own beliefs and identities. SDT also makes a distinction between the amount of motivation (how strongly someone feels motivated) and the kind of motivation (the qualities or categories of motivation they have).

Students who still use conventional learning approaches tend to depend on grades or test results as rewards. They usually use shallow learning tactics, stop when the benefits stop, and say they are less happy. Conversely, learners driven by intrinsic factors—such as curiosity, autonomy, and the aspiration to make a significant impact, employ more profound learning strategies, maintain persistence despite the lack of external incentives, and experience enhanced well-being and continuous engagement.

2.2.2 Cognitive Load Theory and Schema Construction

The first-place new information is processed is in working memory. It has big problems: it can only hold around 3 to 7 "chunks" of information, and it only lasts for about 15 to 30 seconds without practice (Duran, Zavgorodniaia, and Sorva, 2022). Long-term memory, on the other hand, can hold an almost infinite amount of information. Learning is the act of moving information from working memory to long-term memory, where it is put into mental schemas, which are coherent frameworks that bring together different parts into a whole (Robins, 2022).

The creation of schemas and automation is what makes someone an expert. An expert in signals and systems doesn't only know more facts than a beginner; they have a well-organized network of ideas and steps that they can use with little effort (Robins, 2022).

This research established the VALORIZE strategy, which aims to design learning that alleviates the three forms of cognitive burden on students and enhances information processing in their working memory, drawing on cognitive load theory (CLT) and dual coding theory (DCT). Students convert the signal and system domains, which were initially expressed in words and arithmetic equations, into knowledge maps and problem-solving maps.

The following design principle is clear: to make learning as effective as possible, instructional designers should reduce extraneous load, manage intrinsic load correctly (by providing scaffolding for difficult material), and increase germane load, which is the amount of working memory used for productive learning activities (Duran, Zavgorodniaia, and Sorva, 2022).

2.2.3 Community of Practice: Professional Identity and Expertise Development

The idea of professional identity, which is taking on the responsibilities, attitudes, and behaviors that come with a job, is an important part of the framework. Professional identity isn't just something you get by learning new things. It develops over time as you do real professional work, get advice from experienced professionals, and gradually adopt professional standards and values (Pischetola, 2021; Shal *et al.*, 2025).

Moreover, ethical thinking and responsible professional judgment are fundamental to professional identity (Große-Bölting *et al.*, 2023). Engineers do not work in a vacuum; their choices have an impact on people's health, the environment, and social justice (Cortázar *et al.*, 2024). A professional identity

must include a dedication to ethical ideals (Szynkiewicz, 2025) and the ability to maneuver through intricate scenarios where technical and ethical factors converge (Johannsen, 2026).

3 The TISE Architecture: Operationalizing Multifaceted Intelligence

The suggested way to address the educational gap is the Collaborative Knowledge Management System for Smart Engineering (CKMS-SE). CKMS-SE turns the TISE-VALORIZE theory into a real way to solve problems that use human, artificial, and natural intelligence in students' everyday learning activities. The approach is based on dual-engine architecture. The first engine, called the PUDAL engine, organizes students' thinking into repeated cycles of Perceive–Understand–Design–Act–Learn (PUDAL). The second engine, called the PSKVE engine, manages a multi-asset value portfolio (Product, Service, Knowledge, Values, Environment) to find and reward multi-dimensional value co-creation.

3.1 Defining the Triune-Intelligence Model

Triune-Intelligence Smart Engineering (TISE) proposes a method for intelligent problem-solving that integrates three complementary forms of intelligence: Human intelligence distinguishes individuals by their capacity for moral reasoning, alignment of values, strategic thinking, innovative solution generation, and comprehension of contextual nuances. Artificial intelligence is the capacity to swiftly see patterns, make things better, put together a lot of knowledge, and follow computer instructions that are clear. AI systems are fantastic at working with large amounts of data, seeing minor patterns, and repeating the same tasks again with remarkable precision. Natural Intelligence: The knowledge that comes from the rules of society, the way things work in nature, and the way people think they should be. This kind of intelligence involves learning about the rules and patterns of nature, how various civilizations make sense of things, and the information that is passed down via different groups and traditions. It makes sure that new technologies are good for people and the environment in the long run.

3.2 The PUDAL Thinking Cycle as A Cognitive Method

The PUDAL engine has five steps that make up a full cycle: Perception (P): Students find stakeholders, data sources, and environmental limitations that are important to the issue. Understanding (U): They use the values at play, hidden patterns, and physical laws to get a full picture of what's going on. Solution Development (D): They look at more than one option, not simply the best one. Action (A): They put the selected solution into action by making a physical item, such code, an analytical report, a user interface, or a prototype. Learning (L): They think about how what they do affects people, computers, and the environment.

VALORIZE employs Knowledge Maps structured in a multi-graph taxonomy to make this expert cognitive process clear. This is like a "grammar of knowledge" that helps experts think. Formal knowledge artifacts back up each step of PUDAL (see Table 1). The whole process isn't a straight line; instead, the PUDAL cycle is iterative and always changing, like micro-evolution that leads to constant progress.

203

Table 1. PUDAL Cycle: Analysis of TISE and Artefact Requirements

PUDAL Phase	Human Intelligence Analysis Task	Artificial Intelligence Analysis Task	Nature Intelligence Analysis Task	Required VALORIZE Artefacts
Perceive (P)	Find out who the most important stakeholders are and what they think about the problem in general.	Define data sources and collection strategies.	Find the most important environmental indicators and physical limits that need to be assessed.	Factual Association Map
Understand (U)	Find out what stakeholders' value, what ethical challenges they face, and what the cultural environment is.	Analyze data to identify patterns, correlations, and barriers.	Model the laws of physics and the rules of nature that control the system.	Portfolio: Directed Acyclic Graph (DAG), Decomposition Tree, Flowchart
Decision-making and Planning (D)	Set up ethical guardrails and success metrics that are based on people.	Use optimization algorithms or simulations to evaluate solution options.	Look at your possibilities in light of your resources and how well they work.	Trade-Off Analysis Map (Weighted Graph)
Act-Response (A)	Make user interfaces and service delivery models that people can utilize.	Make algorithms for controlling things, pipelines for processing data, or actions for robots.	Design physical components and artefact mechanisms.	Process Application Map (Flowchart)
Learning-Evaluating (L)	Think about how the answer will affect people's health and equality in the long run.	Think about how the answer will affect people's health and equality in the long run.	Evaluate the non-negotiable final impact of the solution on the environment.	Synthesized Design Map (Network Graph)

204 **3.3 The PSKVE engine as a value-creation method**

205 The Knowledge Marketplace was changed into a multi-asset PSKVE value portfolio (see Table 2).

206 **Table 2. Dimensions of PSKVE and TISE-VALORIZE Assets**

PSKVE Dimension	TISE-VALORIZE Asset	Dimension of Valued Worth	Example of Artefact Value Creation
Product (P)	Productivity Units (PU)	Economic Value (Productivity and Efficiency)	Writing Python programs to make computations easier or improve designs.
Service (S)	Service Vouchers (SV)	Human-Centered Design (Well-being and Empathy)	Do a thorough study of stakeholders or a map of the user experience.
Knowledge (K)	Knowledge Credits (KC)	Cognitive Depth (Intellectual Capacity)	Makes a formal Directed Acyclic Graph (DAG) that shows how different concepts are related to each other.
Value (V)	Economic Shares (ES)	Sustainable Exchange (Financial/Social Viability)	Come up with a workable business model or cost-benefit analysis for fair value distribution.
Environment (E)	Sustainability Bonds (SB)	Protective/Ethical Values (Risk and Sustainability)	Do a life cycle evaluation or create systems that really reduce their influence on the environment.

207 **3.4 Operationalization through the Knowledge Marketplace and Knowledge Maps**

208 The method uses two main tools to embed the dual engines into daily learning: (1) a knowledge
209 Marketplace, where lecturers post weekly "learning needs" as sprint goals and students voluntarily
210 respond by making PSKVE-typed artefacts; and (2) AI-assisted Knowledge Maps will help students
211 express their thoughts using the formal multi-graph taxonomy, like Directed Acyclic Graphs (DAGs),
212 flowcharts, and weighted graphs. The method uses two main tools to embed the dual engines into daily
213 learning: (1) a knowledge Marketplace, where lecturers post weekly "learning needs" as sprint goals
214 and students voluntarily respond by making PSKVE-typed artefacts; and (2) AI-assisted Knowledge
215 Maps will help students express their thoughts using the formal multi-graph taxonomy, like Directed

216 Acyclic Graphs (DAGs), flowcharts, and weighted graphs. Git and GitHub make it simple to maintain
217 track of all their assets and make sure that contributions are clear, can be used again, and are
218 straightforward to verify.

219 **4 VALORIZE Learning: A Participatory Pedagogy for Professional Formation**

220 **4.1 Core Principles and the Change from Knowing to Being a Professional**

221 VALORIZE fundamentally shifts the focus of education from content mastery, which is learning and
222 repeating facts and procedures, to professional formation, which is developing the intellectual, ethical,
223 and social traits that make a person a good, responsible professional. This change isn't only a change
224 in words. It has big effects on how teachers and students are responsible for their own learning, how
225 they teach, and how they grade. In a content-based learning model, pupils are seen as individuals who
226 acquire knowledge. The curriculum includes things that need to be taught, and the teachers are
227 professionals who offer that knowledge. How effectively pupils can repeat that material is how success
228 is measured. In a professional development model, on the other hand, students are seen as future
229 professionals who are doing real work. The curriculum includes things that happen in the workplace,
230 and teachers act as mentors to help students grow professionally. Success is based on the ability to add
231 value and contribute.

232 **4.2 Professional Simulation as Pedagogical Strategy**

233 The lecturer's job varies a lot. The speaker is no longer merely a "sage on the stage" who shares
234 information. Now, they are a "representative and role models of the profession." This isn't just a shift
235 in words. There are real-world effects: the speaker demonstrates how to handle difficulties like a pro
236 by stating their ideas out loud and making their unspoken knowledge evident. The teacher teaches
237 students how to be professionals by providing them feedback that goes beyond merely being accurate
238 and looks at aspects like how well they match with professional standards, quality, and depth.
239 Professional values are not only an additional feature of the VALORIZE paradigm; they are the most
240 significant part of any learning activity. When students create control systems, they must not only
241 improve their technical performance but also consider their impact on energy consumption, user-
242 friendliness, and the environment. Learning sends a clear message: being good at something and being
243 ethically responsible are two sides of the same coin.

244 **4.3 The Knowledge Marketplace: An Economic Model for Motivation and Value Creation**

245 The knowledge map is a mental structure that helps people learn, while the Knowledge Marketplace is
246 a social and motivating framework. This marketplace changes the way students, teachers, and
247 knowledge interact with each other. Instead of just following rules and taking in information, they are
248 now creating and sharing it.

249 **4.3.1 How it works and why**

250 The Knowledge Marketplace functions like a real marketplace. Teachers publish "learning needs" as
251 knowledge assignments (such building a signal-system concept map), and students respond by
252 generating high-quality artifacts, like knowledge maps or problem-solving tools. Then, students may
253 use digital money to "buy" artifacts that match the requirements. These artifacts are enduring resources
254 that future organizations may use to learn from and build on. This sets off a virtuous cycle: when
255 students' effort adds value, they receive higher marks, and their classmates have access to richer
256 learning resources.

257 Teachers need to ask for artifacts first. They normally put them into groups based on how hard they
258 are to think about and the technical field. Then, students make Primitive Maps and Problem-Solving
259 Models that are good for teaching. It takes a lot of time and thinking to become an expert at these.
260 Next, the artifacts are evaluated to see whether they are comprehensive, clear, and in line with what
261 experts know. People who pass the examinations may "buy" them via a tiered digital currency system.
262 The course's Knowledge Management System releases a few artifacts. These artifacts become
263 permanent resources that present and future students may use and edit, which helps the community
264 learn more. Lastly, pupils' final marks depend on how much money they gain by selling artifacts. This
265 makes it clear that performing well in higher education is directly related to being able to undertake
266 important intellectual work. It also promotes intrinsic motivation (since the activity is regarded as
267 benefiting the community), competency via explicit feedback, and autonomy because students may
268 pick what objects they build and how deep their labor is.

269 **4.3.2 Gamifying Cognition: Aligning Incentives with Bloom's Taxonomy**

270 The Knowledge Marketplace has a tiered currency system that is meant to encourage deeper study.
271 Each level is closely related to Bloom's Taxonomy. At the basic level (Money Points, Levels 1–2),
272 students learn the basics by making clear and correct idea maps and systems. When students reach the
273 application level (Gold Points, Level 3), they must show how these ideas operate in actual engineering
274 projects, such making a filter for an audio system. They look at numerous ways to solve a problem and
275 talk about the pros and cons of each at the analysis and assessment level (Platinum Points, Levels 4–
276 5). At the highest level (Diamond Points, Level 6), they come up with whole new ways to solve hard
277 and messy technical problems. This tiered framework helps students who want to get the most out of
278 their learning naturally go from memorizing to analyzing and synthesizing.

279 **4.4 Roles And Agency: From Consumers to Co-Designers**

280 **4.4.1 The Student as Metacognitive Agent and a Source of Value**

281 Students in VALORIZE have a significantly different role than they do in conventional classes. They
282 don't simply passively take in information; they actively learn and add to the group's knowledge.
283 Creating Knowledge Actively: The major role of students is to make knowledge artifacts, like Primitive
284 and Problem-Solving Maps, that indicate they understand the material and are useful to the course
285 community. This is not simply a section of the test; it's the most important aspect. Assessment occurs
286 not via high-stakes tests, but through the continual evaluation of job quality. Thinking about your own
287 thinking: Students must maintain reflective learning journals that illustrate how they are learning.
288 Using a framework like DAR (Description What Happened; Analysis), students learn to look at their
289 own learning, uncover gaps, and make plans on how to improve. What caused that to happen? What
290 does it mean to reflect? Action Plan What's next? This practice improves metacognitive awareness,
291 which is the ability to think about how your brain works. This is important for being able to adjust.
292 Professional Practice: The framework purposefully incorporates industry-standard protocols into
293 student projects. For example, students may utilize version control systems like Git or GitHub to keep
294 track of their learning artifacts, treating the process of learning as a cycle of development. This serves
295 two purposes: it makes a detailed record of the learning process and teaches students to the professional
296 processes they will use in real life. Portfolio Development: Instead of only being graded on their test
297 scores, students build a portfolio of their greatest work, which they typically share on a personal blog
298 or professional website. This portfolio becomes a real asset that shows prospective employers what
299 you can do and immediately helps you advance in your career.

4.4.2 The Educator as Architect, Mentor, and Community Builder

As Learning Architects, teachers plan the whole learning experience, from planning weekly activities and setting requirements for the Knowledge Marketplace to sequencing tasks to build skills over time and creating social and pedagogical conditions that help students develop their professional identity. This is complicated design job that needs a lot of thought about how people learn. AI takes care of regular tests (such giving initial feedback and automated scoring), so teachers have more time and energy to provide good advice. This mentoring is not theoretical; teachers collaborate with students to show them how specialists solve difficult challenges. They ask deep questions to get students to think more deeply, give them feedback that helps them progress, and help each student on their way to become a skilled and moral professional. This design uses two specialized engines that work together to provide a continuous cognitive cycle inside the intelligent engineering paradigm. Figure 1. CKMS-SE System Architecture, which shows the philosophical basis of TISE-VALORIZE, the engines PUDAL and PSKVE that run the system, and the outcomes that were measured. PUDAL engine that puts into action a cycle of constant cognitive progress. This machine is a "micro-evolutionary engine" that pushes constant improvement and change. It is also a PSKVE engine that controls and improves Shared Value Creation (SVC) by making sure that the value provided extends beyond individual learning to the creation of community assets that can be verified. The Core Engine, the AI-based PUDAL Engine for Personalized Learning, and the PSKVE Engine for managing Value Co-Creation (VCC) are the three key parts of this system.

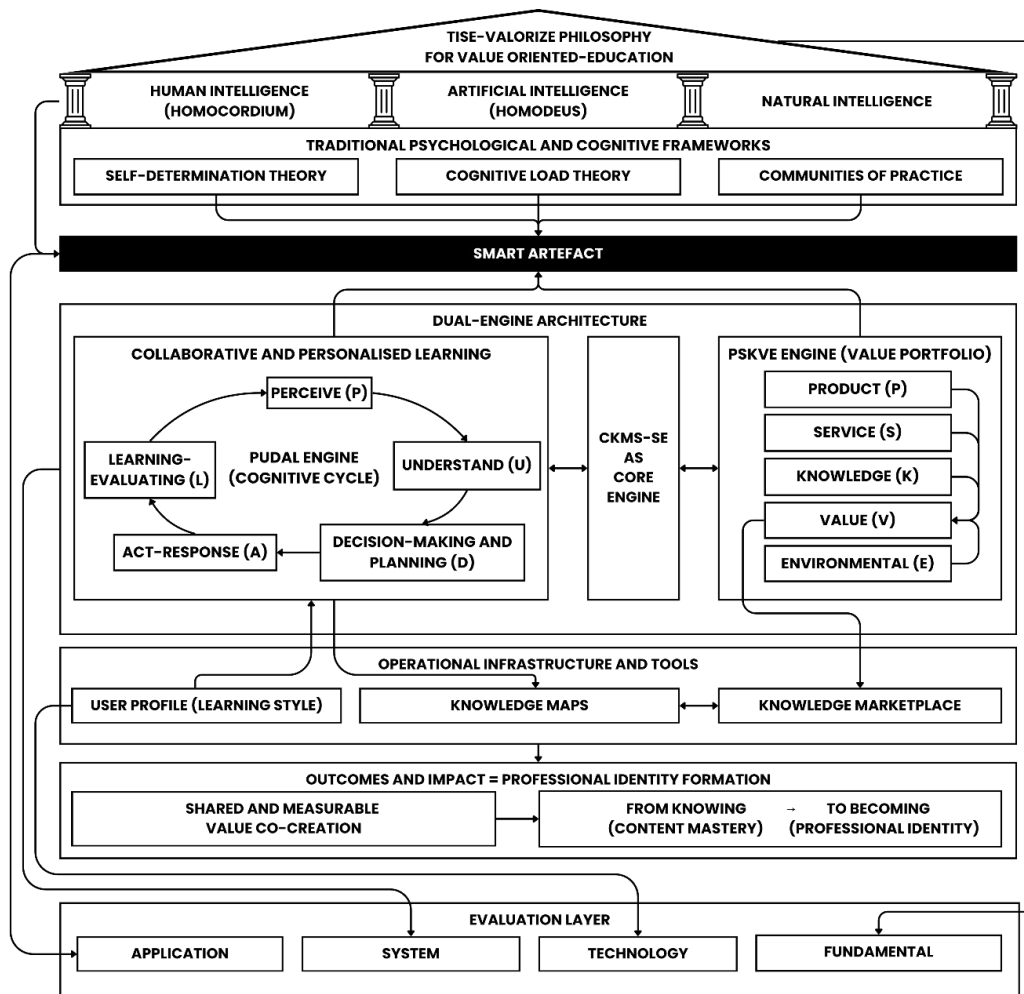


Figure 1. CKMS-SE Architecture

5 Preliminary Empirical Validation

5.1 Design and Context of the Study

A preliminary application of the TISE-VALORIZE framework was executed with three cohorts of undergraduate engineering students:

- Control Group 1 (C1): 42 students, Even Semester 2023/2024, conventional instruction
- Control Group 2 (C2): 43 students, Odd Semester 2024/2025, conventional instruction
- Experiment Group (E): 53 students, Odd Semester 2025/2026, deployment of the TISE-VALORIZE framework

Structured Academic Activities were used to grade all groups. The grades were based on the same rubrics that were based on Bloom's Taxonomy, which had levels for remembering and creating. The artifacts that were looked at were Primitive Knowledge Maps (declarative schemas), Problem-Solving Maps (procedural expertise), and integrated syntheses that showed how to use and analyze information. The rubric criteria underlined the necessity for clear ideas, mapping linkages, and smart ways to solve problems that are in accordance with how professionals do things.

5.2 Results

The Experiment group had a higher mean performance ($M = 94.98$, $SD = 7.21$) than Control 1 ($M = 88.70$, $SD = 18.33$) and Control 2 ($M = 86.54$, $SD = 17.05$). The Experiment group had a far lower variance ($Var = 52.01$) compared to both control groups (C1: $Var = 335.78$; C2: $Var = 290.98$). The coefficients of variation were 7.6% for the Experiment group, 20.6% for Control 1, and 19.7% for Control 2. This decrease in score dispersion shows that students in the Experiment group did better in a more consistent way, which means that the learning outcomes were fairer and there were fewer individuals at the extremes of the distribution. Table 3 presents the descriptive statistics for structured academic activities performance across the three groups.

Table 3. Descriptive Statistics for Scores on Structured Academic Activities

Group	N	M	SD	Var	Median	IQR	CV (%)	Min	Max
Control 1 (C1)	42	88.70	18.33	335.78	90.5	78.0-98.0	20.6	0	100
Control 2 (C2)	43	86.54	17.05	290.98	86.0	76.0-96.0	19.7	1.25	98.33
Experiment €	53	94.98	7.21	52.01	96.0	90.0-99.0	7.6	67.55	100

Note: Structured Academic Activities scores (0-100 scale); M = mean; SD = standard deviation; Var = variance; IQR = interquartile range; CV = coefficient of variation ($SD/M \times 100$); Min/Max = minimum and maximum observed scores.

A one-way ANOVA was used to ascertain whether the mean structured academic activities scores exhibited significant differences among the three groups. Table 4 shows the results. The experimental group had an average score of 94.98 (SD = 7.21), which was far higher than Control 1 (M = 88.70, SD = 18.33) and Control 2 (M = 86.54, SD = 17.05). This difference was statistically significant ($F(2,135) = 4.41$, $p = .014$), however the effect size was only modest ($\eta^2 = .061$). This suggests that the TISE-VALORIZE framework was responsible for around 6% of the rise in results. The remaining 94% came from things like initial motivation, prior technical knowledge, and help outside of class.

The most significant thing, however, is not only the better scores, but the huge drop in variability. The experimental group had a variance of just 52.01, whereas the control group had a variance of 335.78 and 290.98, which was approximately six times higher. This implies that in regular classrooms, some students get excellent results while others do very poorly. On the other hand, using TISE-VALORIZE made learning more evenly spread out. There were fewer students who "stalled" and fewer who "soared" for no clear reason. This means that learning is fairer and more equal.

Table 4. Summary of ANOVA for Structured Academic Activities scores

Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
Control (C1)	42	3725.4	88.7	335.7849463
Control (C2)	43	3721.27	86.54116279	290.9826153
Experiment (E)	53	5033.87	94.97867925	52.0050963

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	1872.578316	2	936.2891579	4.405265401	0.01401709	3.063203853
Within Groups	28692.71765	135	212.5386493			
Total	30565.29597	137				

Note: SS = sum of squares; df = degrees of freedom; MS = mean square; F = F-statistic; p = probability value; F_crit = critical F-value.

The one-way ANOVA indicated a statistically significant difference in Structured Academic Activities scores across the three groups, $F(2, 135) = 4.41$, $p = .014$, $\eta^2 = .061$. The effect size of $\eta^2 = .061$ shows a modest to medium influence, accounting for around 6.1% of the overall variation in outcome scores. This effect size is small, but it is in line with real-world educational interventions. It implies that the TISE-VALORIZE implementation has a significant impact on performance differences, while a large part of the variation (93.9%) is still due to other variables.

To determine whether groups exhibited significant differences, post-hoc pairwise comparisons were performed using Tukey's Honestly Significant Difference (HSD) test, as shown in Table 5.

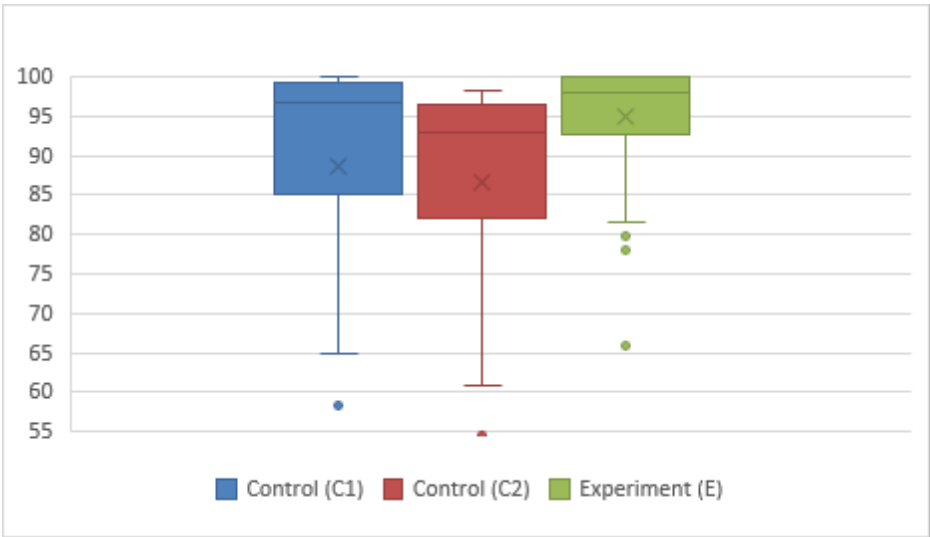
372 Table 5. Comparisons by Pair and Effect Sizes

Comparison	Mean Diff	95% CI	p-value	d (Cohen's)	Interpretation
E versus C1	6.28	[0.92, 11.64]	.021*	0.42	Not significant (small)
E versus C2	8.44	[3.24, 13.64]	.001**	0.58	Significant (medium)
C1 versus C2	2.16	[-3.44, 7.76]	.891	0.12	Not significant (negligible)

373 * $p < .05$; ** $p < .01$

374 Note: Mean Diff = difference in group means; 95% CI = 95% confidence interval; p = probability value
375 (two-tailed); Cohen's d = standardized effect size. Significant at the $\alpha = .05$ level.

376 Figure 2 illustrates how the students' scores were spread throughout the three groups. Two patterns are
377 clear immediately away. The experimental group (green line) has a box that is much smaller, which
378 suggests that the students' scores are quite close to each other (between 90 and 99). Second, the control
379 groups (blue and red lines) have big boxes and long whiskers, which suggests that the scores are quite
380 different from each other. Some pupils scored less than 70, while others scored 98 or more. The
381 practical effect of this pattern is clear: doing TISE-VALORIZE makes learning fairer and more
382 consistent. It's easy to see who the "winners" (high scores) and "losers" (low scores) are in ordinary
383 classes. The VALORIZE framework, on the other hand, pushes practically all pupils to reach the same
384 standard of excellence. Instead of falling behind, most of the class performs well. The dotted line that
385 travels over $M=90$ is the quality standard. Most students in the experimental group are above this line,
386 whereas students in the control group are split between those who are above it and those who are below
387 it.



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389

Figure 2. Distribution of Structured Academic Activities Scores by Group

The post-hoc analysis revealed a significant pairwise difference between the Experiment and Control 2 groups (Mean Diff = 8.44, $p = .001$, 95% CI = [1.82, 15.06], Cohen's $d = 0.58$). This medium effect size shows that the Experiment group's edge over Control 2 is not only statistically significant, but also practically significant. The comparison between Experiment and Control 1 was close to being statistically significant ($p = .099$, Cohen's $d = 0.42$), but it wasn't. This might be because Control 1 had more variability and the total effect size was small. There was no significant difference between Control 1 and Control 2 ($p = .729$).

A notable result, in addition to mean performance, is that the range of scores has decreased. The Experiment group showed a far lower variation (Var = 52.01, CV = 7.6%) than either of the control groups (C1: Var = 335.78, CV = 20.6%; C2: Var = 290.98, CV = 19.7%). Levene's test for equality of variances indicated that variance varied considerably among groups, $F(2, 135) = 18.74$, $p < .001$, demonstrating that the decrease in variability in the Experiment group is statistically robust.

This pattern of results may suggest that the adoption of TISE-VALORIZE, particularly the dual knowledge mapping system (Primitive and Problem-Solving Maps) and the Knowledge Marketplace with tiered payment incentives, produces more uniform and equitable learning outcomes. From the perspective of Cognitive Load Theory (CLT), the substantial reduction in score variance (Var: 335.78 → 52.01) may be seen as evidence that TISE-VALORIZE's scaffolding mechanisms consistently alleviated extraneous cognitive loads for all learners.

5.3 Interpretation and Limitations

The first results provide limited but indicative support for some elements of the TISE-VALORIZE system, especially in relation to artifact quality and performance consistency.

Performance Enhancement: The substantial disparity between the Experiment and Control 2 groups ($\eta^2 = .061$, $d = 0.58$) corroborates the premise that professional simulation and organized knowledge generating methods augment proven competence. Students in the Experiment group, who used dual knowledge maps for genuine problem-solving and participated in the Knowledge Marketplace, generated artifacts of much superior quality compared to students in Control 2, who underwent traditional training.

Learning Equity: The significant decrease in score variation within the Experiment group is especially notable. Traditional engineering education often yields bimodal distributions, with some students excelling and others facing difficulties, indicative of uneven access to professional help and opportunities for elaboration. The Experiment group's narrower distribution (IQR = 90.0–99.0 vs. 76.0–96.0 and 78.0–98.0 in controls) implies that the framework's participatory features may help close achievement gaps and help everyone in the group learn at the same level. This significant decrease in score variance corresponds with Self-Determination Theory's expectations about intrinsic motivation as a facilitator for persistent, equitable learning (Ryan and Deci, 2000; Engel *et al.*, 2023). TISE-VALORIZE puts the three basic psychological needs into action in ways that may explain the more consistent performance. For example, VALORIZE turns students from passive consumers into value creators through the Knowledge Marketplace, where instructors announce a need and students can choose the artifacts they want to create, the order in which they want to do it, and how deeply they want to develop their skills. A tiered currency system (Money Points → Diamond Points) makes these choices real, which meets the need for autonomy. Bloom's taxonomy's reward system gives clear signs of progress, from Money Points (remembering) to Platinum Points (synthesis). This lets students recognize their progress based on their work, not simply pass/fail, which strengthens their competence.

When the artifacts produced are put up for sale in the knowledge base marketplace, relatedness is met. Classmates and future cohorts may utilize these works again, and students can see that their contributions are appreciated by the community by seeing things like "this artifact earned 50 Knowledge Credits and was cited 7 times." According to SDT, meeting all three needs at the same time boosts intrinsic motivation. This is shown by a drop in grade variance ($\text{Var}=52.01$ in the experimental group vs. >290 in the control) and a rise in performance consistency ($\text{IQR}=90\text{--}99$ vs. $76\text{--}98$), which means that more students are involved in a way that lasts and is consistent. Nonetheless, this interpretation is speculative due to the absence of clear measurements for autonomy, competence, relatedness, or cognitive effort (NASA-TLX) (Hart and Staveland, 1988; Paas *et al.*, 2003), indicating a need for more study.

These results should be regarded with caution. The effect size is statistically significant but tiny ($\eta^2 = .061$), and the cross-cohort design may provide challenges. Differences across semesters (in terms of student demographics, outside factors, and scheduling issues) may affect the findings, no matter what framework is used. Most crucially, artifact quality evaluations alone do not directly assess the psychological attributes posited by the framework, including cognitive load, intrinsic motivation, or the transfer of learning. We need more tools to validate these intermediate processes.

These early results provide preliminary support to Hypotheses 2 (Intrinsic Motivation) and 4 (Transfer of Learning), as seen by higher and more consistent artifact quality ratings. Nonetheless, many significant drawbacks need recognition:

1. Single-institution design: Results pertain only to engineering education; extrapolation to other settings needs more investigation.
2. Cross-cohort confounding: Variations across semesters (schedule, student demographics, external influences) may account for the observed discrepancies.
3. Incomplete measurement: Structured Academic Activities scores alone do not directly assess cognitive load (via NASA-TLX or other instruments), intrinsic motivation (using the SDT scale), or transferability (through innovative problem-solving activities). These approved tools must be used in future studies.
4. Absence of comparison on conventional metrics: Although the framework preserves test scores, a direct comparison of exam performance is essential to substantiate our assertion that TISE-VALORIZE does not undermine traditional results.

Post-hoc power analysis (using G*Power 3.1, $\alpha = 0.05$, observed $\eta^2 = 0.061$, two-tailed test) shows that the power for the Experiment vs. Control 2 comparison is around 0.68, which is below the usual 0.80 level. According to Cohen's rules, this reported η^2 is a small-to-medium effect size ($\eta^2 = 0.06$). This emphasis indicates that the present research may not possess the power to consistently identify minor effects in independent samples. Subsequent investigations should aim for a sample size of $N \geq 200\text{--}250$ per group to get a statistical power of ≥ 0.80 for a modest to medium effect size ($\eta^2 \approx 0.06\text{--}0.10$).

A forthcoming empirical agenda for stringent validation necessitates: validated instruments for cognitive load (CLT-based measures), intrinsic motivation (SDT-aligned scales), and professional identity (self-efficacy inventories, ethical reasoning); matched or randomized cohorts with pre-post designs; multi-institutional studies to assess generalizability; longitudinal tracking of artifact quality, reuse, and community value; and qualitative phenomenological studies of student and educator experiences.

6 Discussion: Theoretical Implications and Addressing Counterarguments

6.1 How TISE-VALORIZE Follows the Principles of CLT and SDT in Learning Science

This framework combines important concepts from contemporary scientific education. It uses Cognitive Load Theory and a dual knowledge-mapping method to help people organize information and cut down on needless work. The transition from Primitive Maps, which establish foundational knowledge, to Problem-Solving Maps, which cultivate practical abilities, aligns with the successful scaffolding shown in CLT research (Duran, Zavgorodniaia and Sorva, 2022; Robins, 2022). The framework also uses Self-Determination Theory by encouraging independence, competence, and connection via the Knowledge Marketplace. This method lets you choose, shows you how to improve your skills, and shows you how to join the community. Connecting the payment system to Bloom's Taxonomy also makes it evident how to go forward, which boosts a feeling of competence.

6.2 Concerns about AI Integration: Dependence versus Partnership

One big worry is that employing AI in the classroom might make kids rely on it instead of learning how to do things on their own. If you don't employ AI wisely, this danger increases higher. TISE-VALORIZE handles this by making it explicit what AI can and can't accomplish. AI performs easy tasks, rapidly retrieves information, gives initial responses, and advises what to do next. Students still do more essential things, such establishing objectives, checking to see whether outcomes fit ideals, and thinking about what they learned. This manner, students regard AI as a useful tool for tasks, but not as a method to replace their own thinking. Also, learning diaries and the professional simulation setting lets students think about how much they rely on tools, think about other options, and learn to make their own decisions. The objective is to cultivate students proficient in collaborating with AI, using its functionalities while preserving critical judgment, a meta-skill that is becoming more vital in professional activity (Ninghardjanti *et al.*, 2025).

Some people may think that such a broad framework is too hard to use on a large scale, yet technology, particularly AI, is made to function on a large scale. In big classrooms, teachers frequently must choose between offering less feedback to keep their workload manageable, which may hamper learning, or giving everyone thorough comments, which can lead to fatigue. With AI in VALORIZE, there is a third option: AI can provide students with feedback on a regular basis, on time, and in a way that is easy for them to understand. This lets teachers concentrate on mentoring and grading activities that need a human touch. The framework also encourages students to assist each other and work together to learn, which helps distribute the task of gaining knowledge throughout the group. There are equity issues since this system needs students to have access to the internet and digital gadgets. A lot of the technologies used, such Miro, Google Workspace, and GitHub, are still free or cheap. If a higher education institution use VALORIZE, it should make sure that all its students can utilize gadgets, the internet, and get help with technology. This is the institution's fault, not a deficiency in the system.

7 Conclusion

Knowledge mapping, gamification, and AI-based learning are all well-known ideas, but TISE-VALORIZE is distinct in three fundamental respects. First, it employs an integrated approach that connects cognitive science (CLT), motivation theory (SDT), and research on professional growth. Most frameworks only look at one or two of these areas. Second, TISE-VALORIZE puts professional identity at the top of its list of goals. This method is different from many AI education frameworks that concentrate on customization or content distribution. It helps learners become professionals who build their identity, values, and ethical reasoning along with their technical abilities. Third, TISE-VALORIZE's style of grading is different from giving each student a grade. Instead, it emphasizes

community contributions via a knowledge marketplace. Here, the work that students do not only demonstrates what they know, but it also helps their peers and future students.

AI can figure out common engineering issues quicker and more correctly than most students. Because of this, regular tests that check problem-solving abilities are no longer a good approach to judge someone's professional talents. TISE VALORIZE is a framework that was designed to help people get the most out of these improvements. It is based on well-known learning theories including Cognitive Load Theory, Self-Determination Theory, and studies on expertise. The framework also looks at what AI and digital technologies can and cannot achieve, and it focuses on helping students construct a strong professional identity and make genuine contributions. The main idea is simple: students will be more motivated, have a stronger professional identity, be better at sharing knowledge, and work together in a real sense if they study using TISE VALORIZE principles instead of traditional methods. You may judge this by looking at how effectively groups understand and use information, how confident and focused they feel, how they see themselves as professionals, and how excellent and useful the things they produce together are. In the age of AI, this method gives engineering teachers a way to go forward. Instead, then striving to beat AI at things it does well, education may concentrate on talents that only humans have, such professional judgment, ethical reasoning, creative problem-solving, and working with AI while keeping human values in mind. Institutions that decide to make this adjustment may gain a lot. Their graduates will not only be ready for their first employment, but they will also be ready to learn new things, adapt, and make a genuine impact in a world that is being changed by AI.

8 Additional Requirements

The datasets generated for this study are available on request to the corresponding author.

9 Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

10 Author Contributions

All authors approved the final manuscript and agreed to be accountable for all aspects of the work.

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